

2.5 Integration using Tables of Integrals & Computer Algebra Systems

Integration using Tables

The front and back cover of the textbook have tables of integrals. These tables show antiderivatives for more complicated integrals and can be useful at times.

Example 1: Entry number 89 in the table of integrals in your book is:

$$\int \frac{u^2}{\sqrt{a^2 - u^2}} du = -\frac{u}{2} \sqrt{a^2 - u^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{u}{a}\right) + C. \text{ Use this formula to find } \int \frac{x^2}{\sqrt{5 - 4x^2}} dx.$$

Example 2 (you try): The following are entries which appear in the integral table in your book:

$$108. \int \frac{du}{u\sqrt{a+bu}} = \frac{1}{\sqrt{a}} \ln \left| \frac{\sqrt{a+bu} - \sqrt{a}}{\sqrt{a+bu} + \sqrt{a}} \right| + C \text{ if } a > 0$$

$$106. \int \frac{u du}{\sqrt{a+bu}} = \frac{2}{3b^2} (bu - 2a)\sqrt{a+bu} + C$$

Use one of these to find: $\int \frac{\sin(2\theta)}{\sqrt{5 - \sin(\theta)}} d\theta.$

NOTE: For class work, the table of integrals in the textbook should only be used when asked.

Computer Algebra Systems (CAS)

Computer algebra systems can integrate a wide variety of integrals. Computer algebra systems are becoming more predominant especially with free computer algebra systems like Wolfram Alpha becoming available. In this class, since Wolfram Alpha is readily available, this is what we will use.

Example 3: Use a CAS to find $\int \frac{4^x + 10^x}{2^x} dx$.

Example 4: Use a CAS to find $\int \frac{\sin(2\theta)}{\sqrt{5 - \sin(\theta)}} d\theta$.

Even more Complicated Indefinite Integrals!

Example 5: Consider $\int e^{-x^2} dx$.